

Bishal Dhungana

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SUMMARY

Geospatial professional with hands-on experience across the full GIS stack: enterprise application development, spatial database administration, geoprocessing automation, and data quality assurance for critical utility infrastructure. I build and deploy GIS applications using Esri technologies, manage enterprise geodatabases, and work hands-on with ArcGIS Utility Network for electric utility asset and network modeling.

SKILLS

Geospatial: ArcGIS Pro / Enterprise / Online, Utility Network, PostGIS, DuckDB Spatial, GDAL, PDAL, GeoPandas, Shapely, PyProj, LiDAR / Point Cloud, Global Mapper, LAsTools, Overture Maps, Cloud-Optimized GeoTIFF, Folium, FME

Data Engineering: Python, SQL, ETL/ELT Pipelines, PySpark, DuckDB, Databricks, Data Lakes, Parquet

Cloud & DevOps: AWS Lambda, S3, SageMaker, ECR / ECS, AWS CDK, Docker, IAM, VPC / Network Security, CI/CD

Generative AI & Agentic Systems: LangChain, LangGraph, RAG, Model Context Protocol, FastMCP, Vector Databases, Ollama, Chroma, Hugging Face Transformers

EXPERIENCE

Data Scientist (Geospatial & AI)

Apr 2025 – Aug 2026

Florida Power & Light - Contract, West Palm Beach, FL

- Built and deployed GIS applications (ArcGIS Enterprise, Experience Builder, Dashboards, Utility Network) for infrastructure monitoring and disaster response, including spatial web portals used in real time during Hurricane Milton for structural impacts, outages, and crew deployment.
- Built a storm-pilot deployment application in ArcGIS Experience Builder to manage internal and external UAV pilots, with real-time location tracking, multi-deployment coordination, time-of-deployment logging, and multi-level access for enterprise users.
- Designed the annotation-storage schema for multiple model versions, and set up the taxonomy as a feature service with editor logging and concurrent multi-user updates, plus a separate read-only REST API for internal teams and vendors, replacing a fragile Excel-based process.

Associate Geospatial Analyst

Sep 2024 – Mar 2025

Florida Power & Light - Contract, Jupiter, FL

- Led QA/QC of LiDAR-derived power grid data for infrastructure resilience work, including pole hardening.
- Managed PostgreSQL/PostGIS databases, optimized Cloud-Optimized GeoTIFFs, and automated QA workflows in FME, ArcPy, and Model Builder to streamline data storage and validation.
- Built a FastAPI sidecar for heavier geoprocessing and format conversion.

Research Assistant

Jan 2023 – Aug 2024

Florida Atlantic University, Boca Raton, FL

- Conducted FEMA hydrologic and hydraulic (H&H) flood risk assessments for 3 Florida communities using Python automation, part of FAU's \$1.7M FEMA-funded CWR3 Watershed Master Planning Initiative to build a statewide planning template.
- Built an automated ArcGIS Pro geoprocessing pipeline, cutting processing time by 50%.

Geomatics Engineer

Dec 2020 – Dec 2022

Freelance Consultant, Kathmandu, Nepal

- Designed municipal geodatabases (land use, soil, cadastral, resource maps) with topology and attribute rules.
- Served as GIS expert for Nepal's Survey Department land-use hand-over program, training 40+ officials in spatial analysis and database management.

EDUCATION

MSc, Geosciences — GIS Concentration — Florida Atlantic University · 2023–2024

BSc, Geomatics Engineering — Tribhuvan University, Nepal · 2016–2021

CERTIFICATIONS & TRAINING

- The Rise of the AI Data Engineer Bootcamp (2026)
- FAA Part 107 — Remote Pilot Certificate
- Machine Learning Specialization — DeepLearning.AI
- Deep Learning Specialization — DeepLearning.AI
- Remote Sensing Image Acquisition & Analysis — UNSW Canberra
- **NASA ARSET Certifications (12):** Applied remote sensing training (2022-2024) in climate, disaster, and environmental risk modeling.
- **Esri Training Certifications (8):** GIS fundamentals, cartography, spatial data science, and imagery analysis.
- GIS Mapping & Spatial Analysis — U. Toronto
- Data Science with R — Johns Hopkins
- GIS Specialization — UC Davis
- Introduction to Databases — Meta

PUBLICATIONS

Dhungana, B.; Liu, W. *Urban–Rural Exposure to Flood Hazard and Social Vulnerability in the Conterminous United States*. ISPRS Int. J. Geo-Inf. 2024, 13, 339.

Mandal, A.K.; Thapa, A.K.; Thapa Chhetri, M.; Dhungana, B.; Bloetscher, F.; Yong, Y.; Su, H. *PyWMP: A Unified Python Framework for Multi-Fidelity 1D, 2D, and Hybrid Event-Based Flood Modeling*. Manuscript in review, DOI pending; ongoing contributor to the open-source PyWMP package.

PROJECTS

Road-Centerline: A published Python package that extracts road centerlines from polygon geometries — worldwide, in any input CRS, in any format GeoPandas can read. CLI and Python API, computed via medial-axis skeletonization. (*Python, GeoPandas, pygeoops, PyProj, PyPI*) [Repo](#)

HazardMCP: An MCP server unifying FEMA, USGS, NOAA, NWS, and NASA hazard data behind a single AI-agent interface — one-call location risk summaries and map rendering, compatible with Claude Desktop. (*Python, FastMCP, Folium, Docker*) [Repo](#)

SurgeExposure: A cloud-native pipeline overlaying NOAA storm-surge and flood-inundation data against Overture Maps building footprints, scoring per-asset flood exposure across 8 coastal regions — with a map and API. (*Python, DuckDB Spatial, FastAPI, PostGIS, AWS*) [Demo](#) | [Repo](#)

fflood-nep: A reproducible flood-response pipeline for Nepal's August 2026 Rasuwa/Bhote Koshi flash flood — runs Sentinel-1 SAR change detection for flood extent, scores building/road/facility exposure by municipality, and tracks a separate InSAR pipeline for landslide/avalanche source detection. Built and shipped inside the event's first week. (*Python, Sentinel-1 SAR, STAC, GDAL, InSAR*) [Demo](#) | [Repo](#)

3DEP LiDAR Terrain Builder: Generates DEM, DSM, hillshade, slope, aspect, and contour products from USGS 3DEP lidar. Draw an area of interest in the browser; it generates a configured local script — no data ever uploads. (*PDAL, GDAL, RichDEM, AWS EPT/S3*) [Demo](#) | [Repo](#)

3d-export: 3D-style land cover export from Sentinel-2 tiles, clipped to any country and automated via GitHub Actions — a modernized take on the rayshader R workflow. [Demo](#) | [Repo](#)

GeoCoder: Converts a spreadsheet of US street addresses into geocoded points and building footprint polygons using free Nominatim and Overpass APIs — as a no-install web app or a scriptable Jupyter notebook. (*GeoPandas, Nominatim / Overpass, Static Web App*) [Demo](#) | [Repo](#)

GTFS Transit Platform: Data-engineering and analytics platform for GTFS transit feeds — headways, stop coverage, and spacing anomalies, explored in an interactive map app. [Demo](#) | [Repo](#)